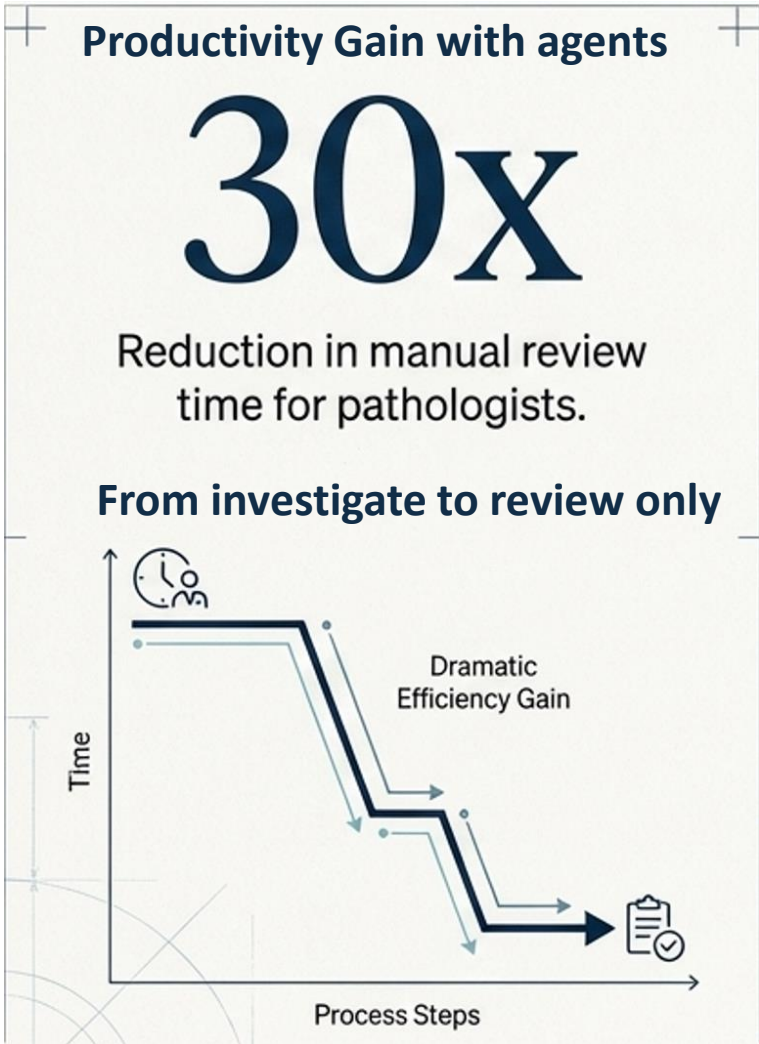
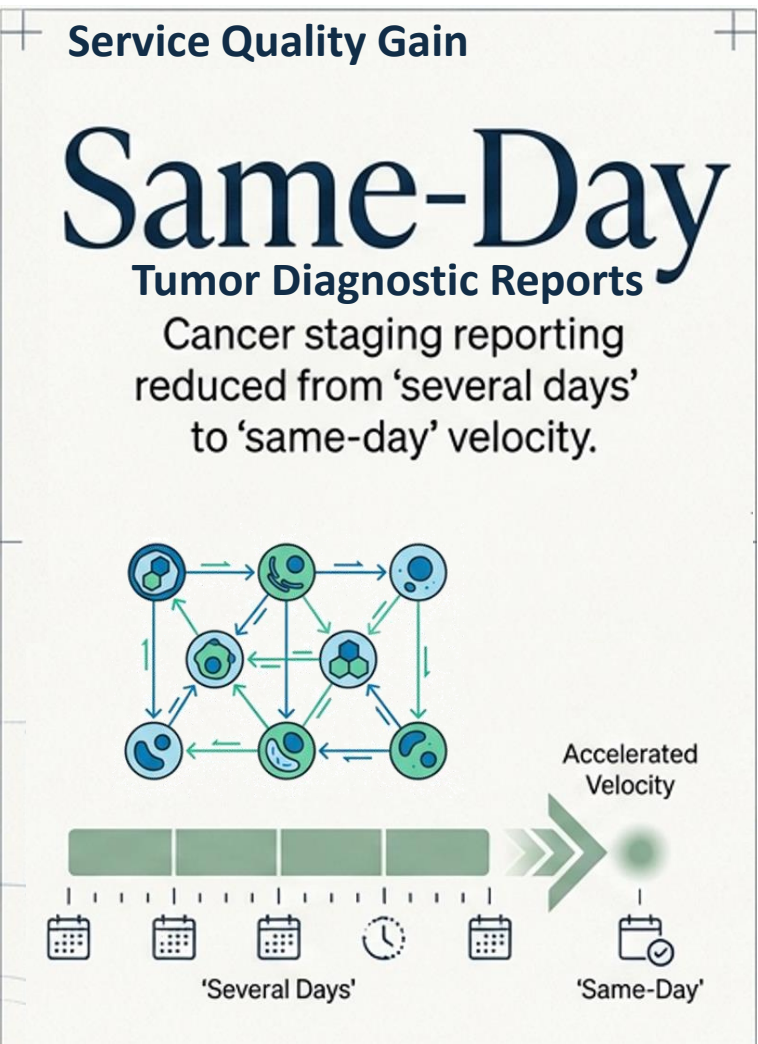
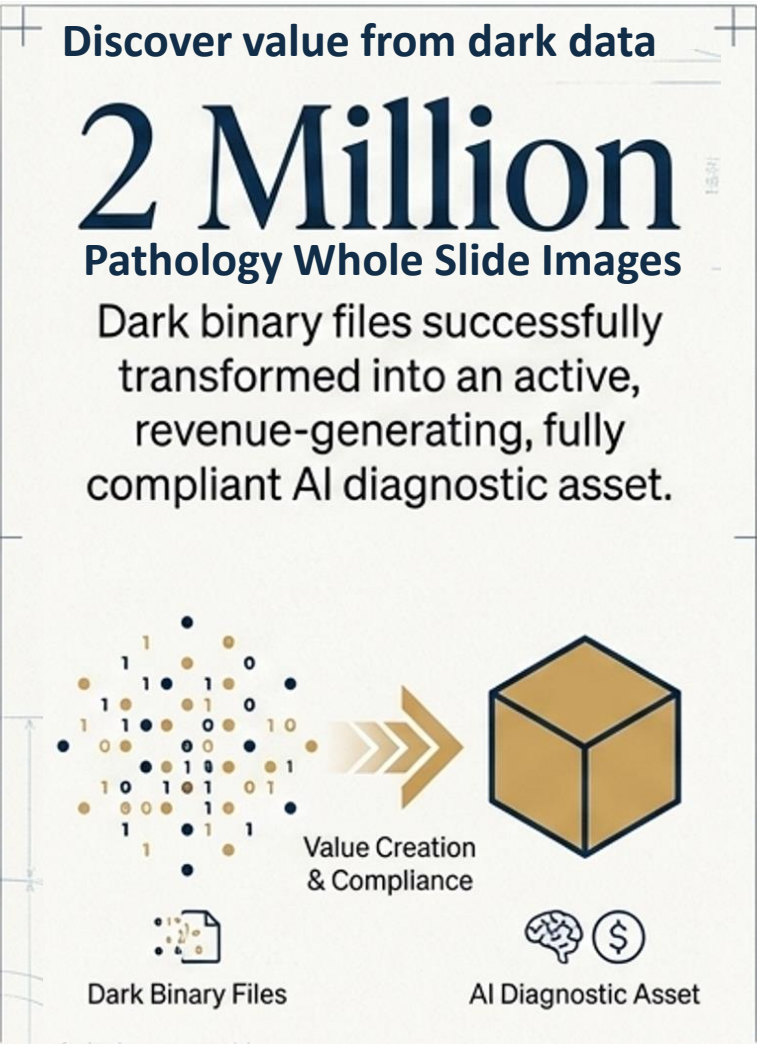


Database Modernization for Digital Pathology with Clinical Foundation Models

This is a story about

Turning dark data from a regulatory liability into an operational asset

Resulting in ...



From Dark Data to Diagnostic Insight:

The RuiPath Transformation at Shanghai Ruijin Hospital

The Challenge: A Legacy of “Dark Data”



Manual Microscope Review

Pathologists were forced to manually review every slide, a process prone to inconsistency and significant diagnostic delays.



The Labeling Bottleneck

The hospital had zero ground-truth labels for tumor stages, making traditional supervised AI training impossible without massive manual effort.

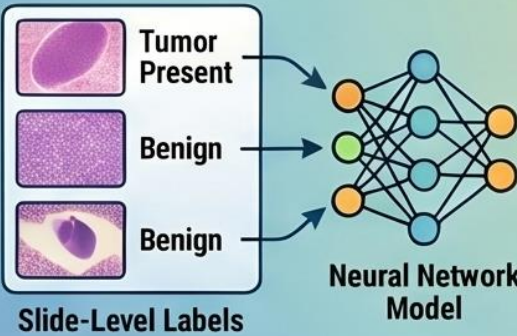
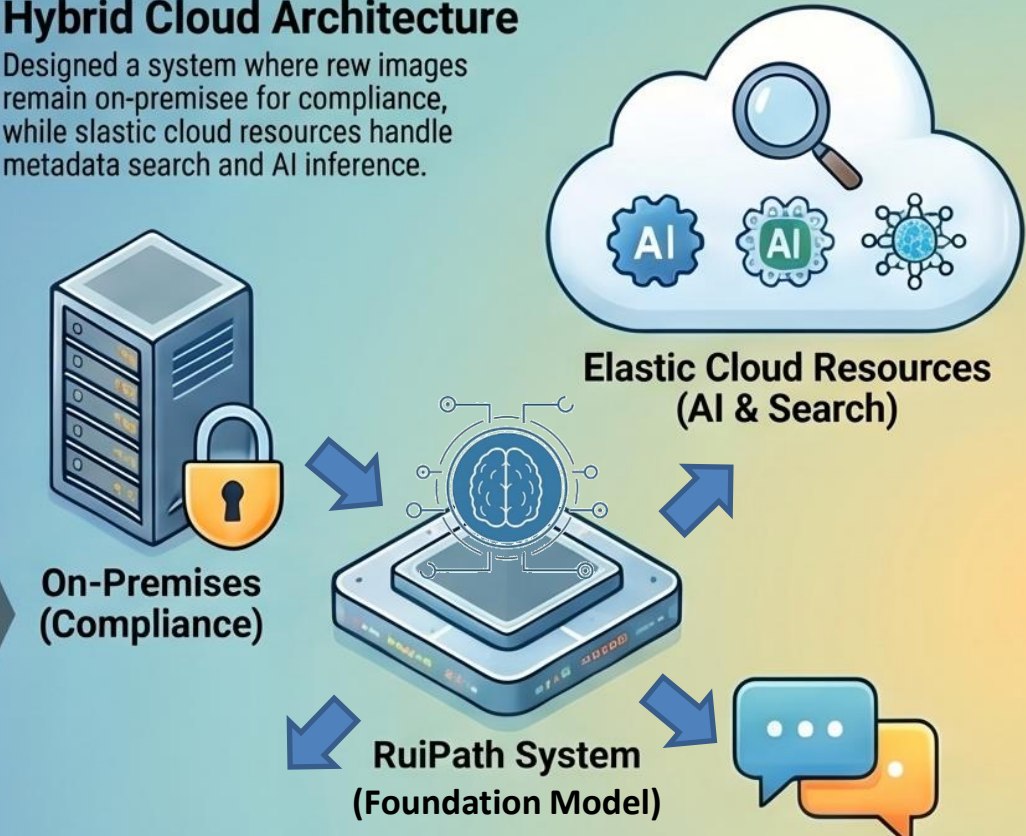
2 Million “Dark” Images

Over a decade of high-resolution whole-slide images were stored as unstructured binary files, inaccessible for AI or machine learning workloads.

The Transformation: Building RuiPath

Hybrid Cloud Architecture

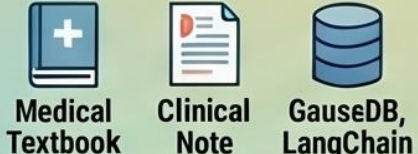
Designed a system where raw images remain on-premise for compliance, while elastic cloud resources handle metadata search and AI inference.



Multi-Instance Learning (MIL)

Implemented “weak supervision” to train models using only slide-level labels from pathologists rather than pixel-level annotations.

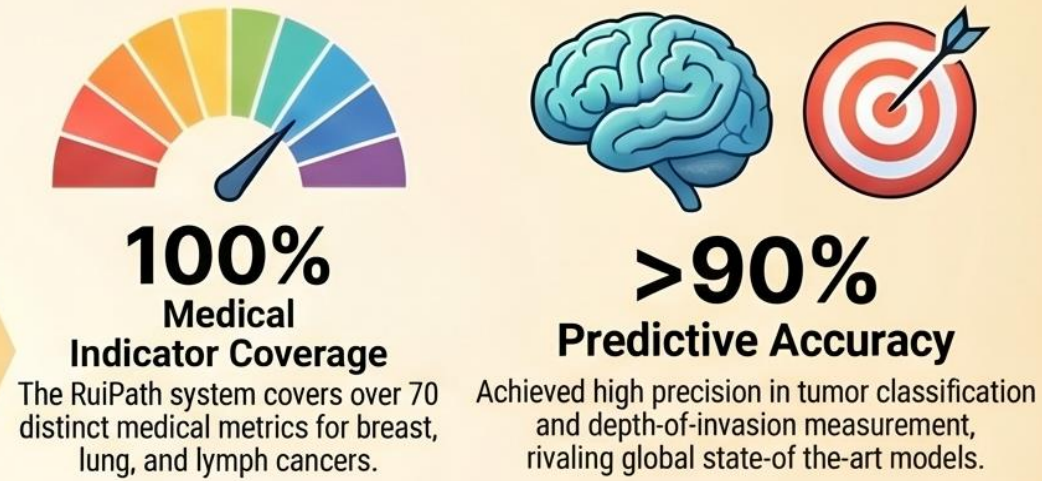
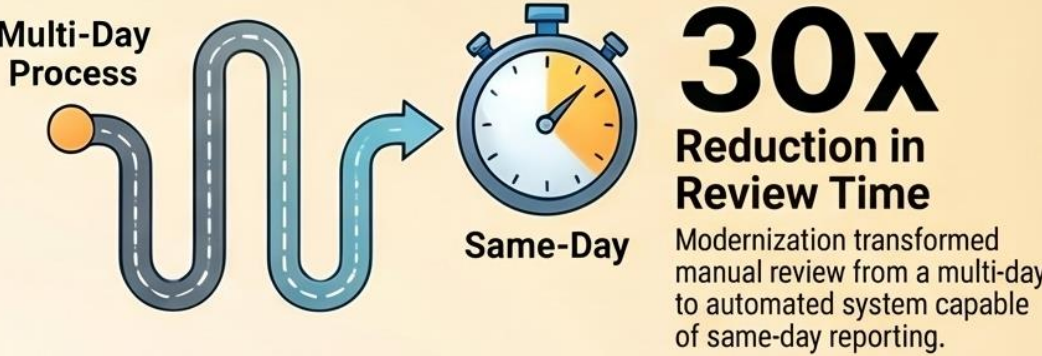
Search



RAG-Based Knowledge Search

Developed a conversational QA system for pathologists using LangChain and GauseDB to query clinical notes and medical textbooks alongside image data.

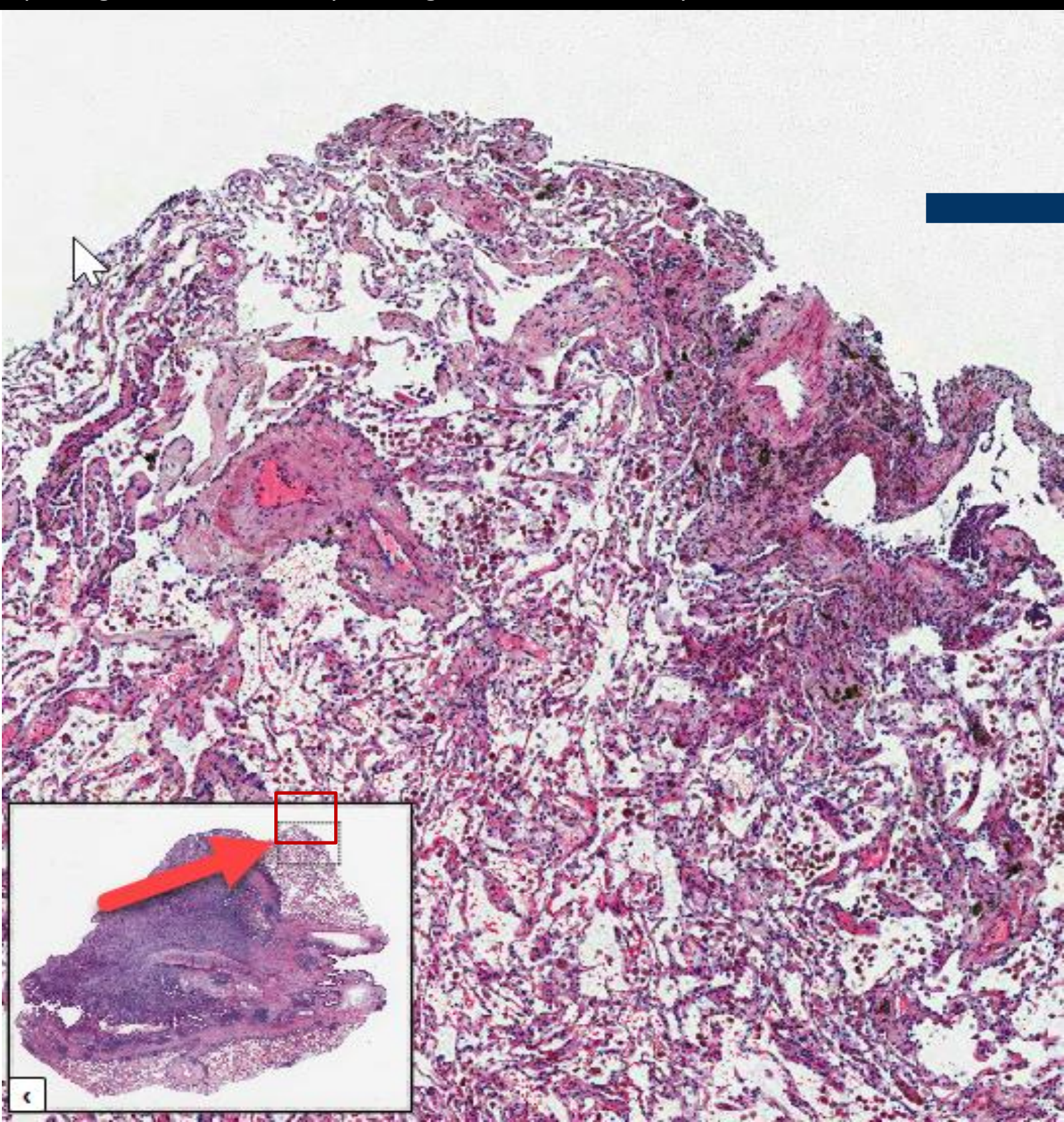
The Value Created: Clinical & Operational Impact



Performance Comparison: RuiPath vs. Global Benchmarks

Task Category	Performance Metric	RuiPath Result	Benchmark Comparison
Clinical Indicators	Coverage Rate	100%	70+ distinct metrics
Tumor Classification	Accuracy	>90%	World-class performance
Research Benchmarks	AUC/ACC/F1	7 Tasks at Global Top	Comparable to Virchow2
Report Turnaround	Cancer Staging	Same-Day	Previously several days

this is a 40x zoom-in image of 1 cm² specimen under digital camera, spotting tumor cells is truly looking for a needle in a haystack, there is AI, but ...

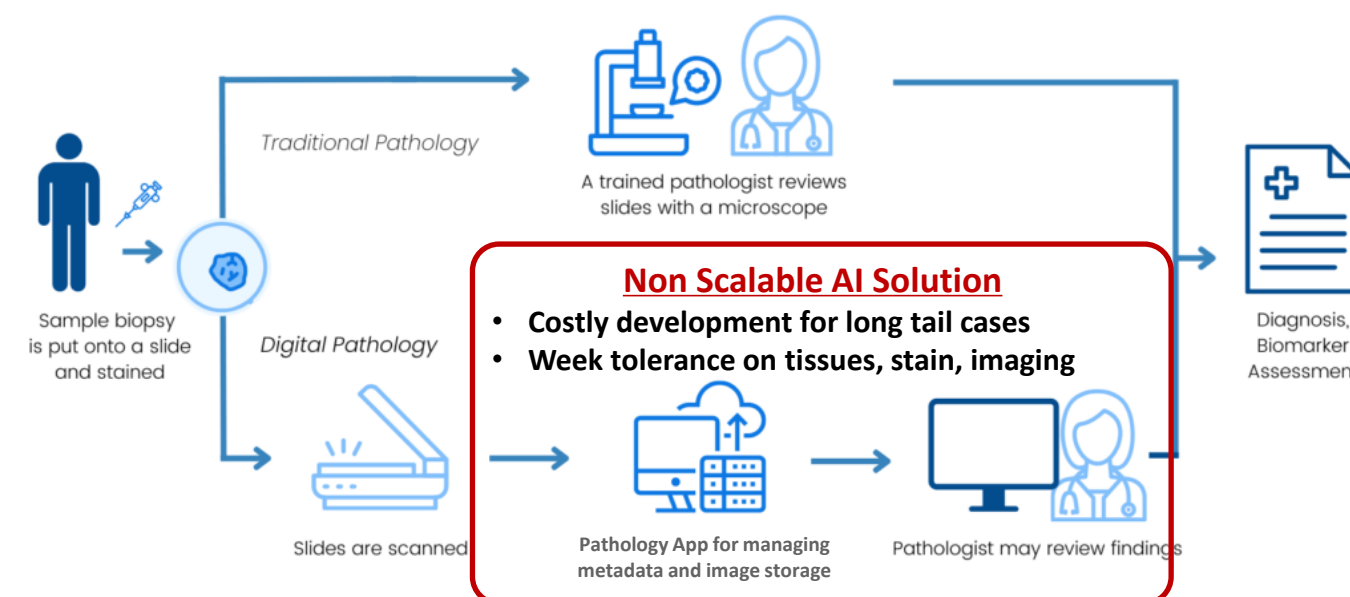


Paint-points

Digital Pathology is NOT Scalable

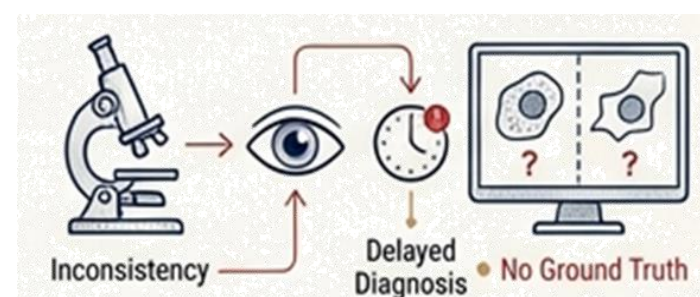
It only create high resolution images, traditional computer vision is **costly** but **weak**

A computer vision model needs thousands of images manually labelled at the cell level for each cancer type and each tissue. Each label costs CNY 2,000 because only a medical specialist can do the work. So, adding just one new feature costs at least CNY 2 million. But unfortunately, there are hundreds features per cancer, so ...



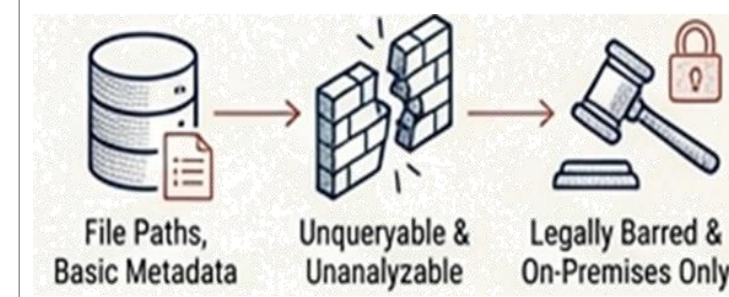
Business Constrains

- 100% reliance on manual review either on big screen or under microscope;
- Tedious, prone to error and inconsistency, cause delay diagnoses.
- Run out of doctors, 10 years to train a certified pathologist (demand > supply)

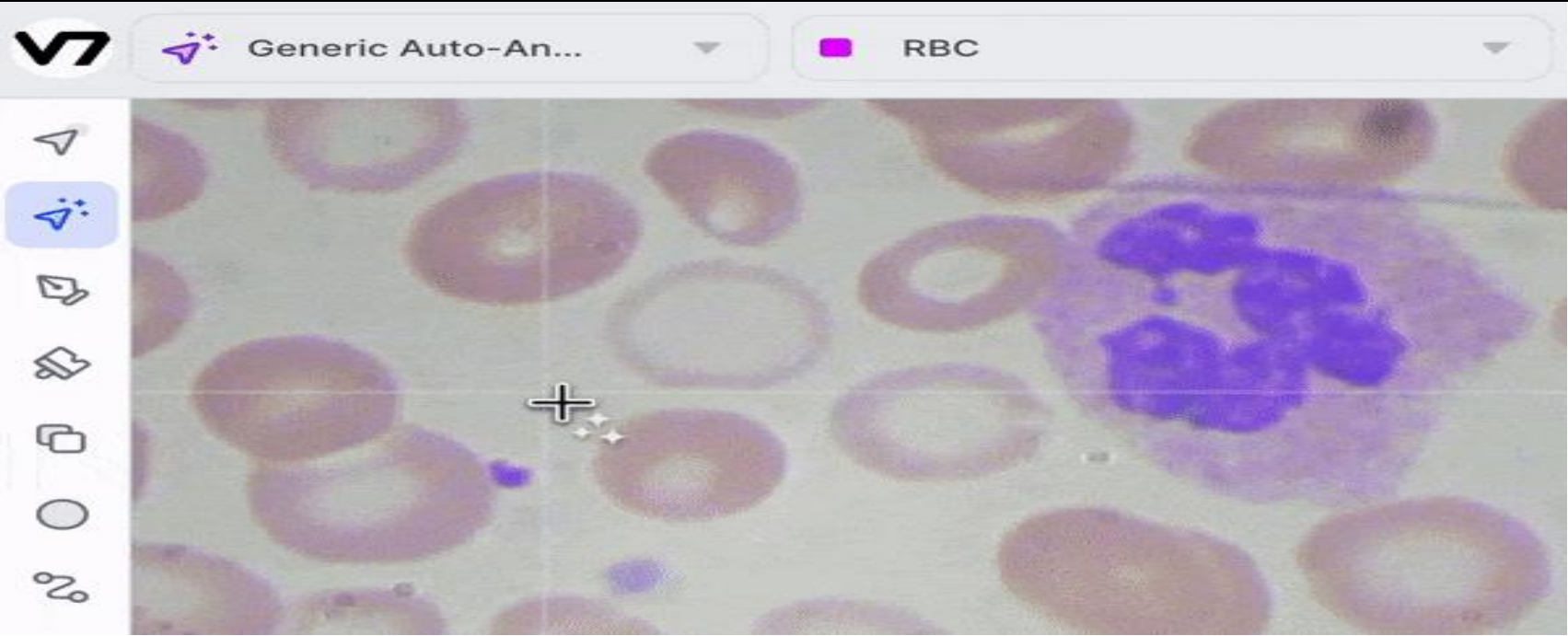


Technology Constrains

- Legacy database stored only metadata and file paths to raw images.
- Extended modules for analytics is week and costly, the hospital just drop those features.
- Data was un-queryable, un-analyzable, and kept for compliance only



Legacy AI (Computer Vision Model) helps doctor performing labeling with intelligent boundary detection, but there are still thousands of those tasks per tissue.



AI Solution

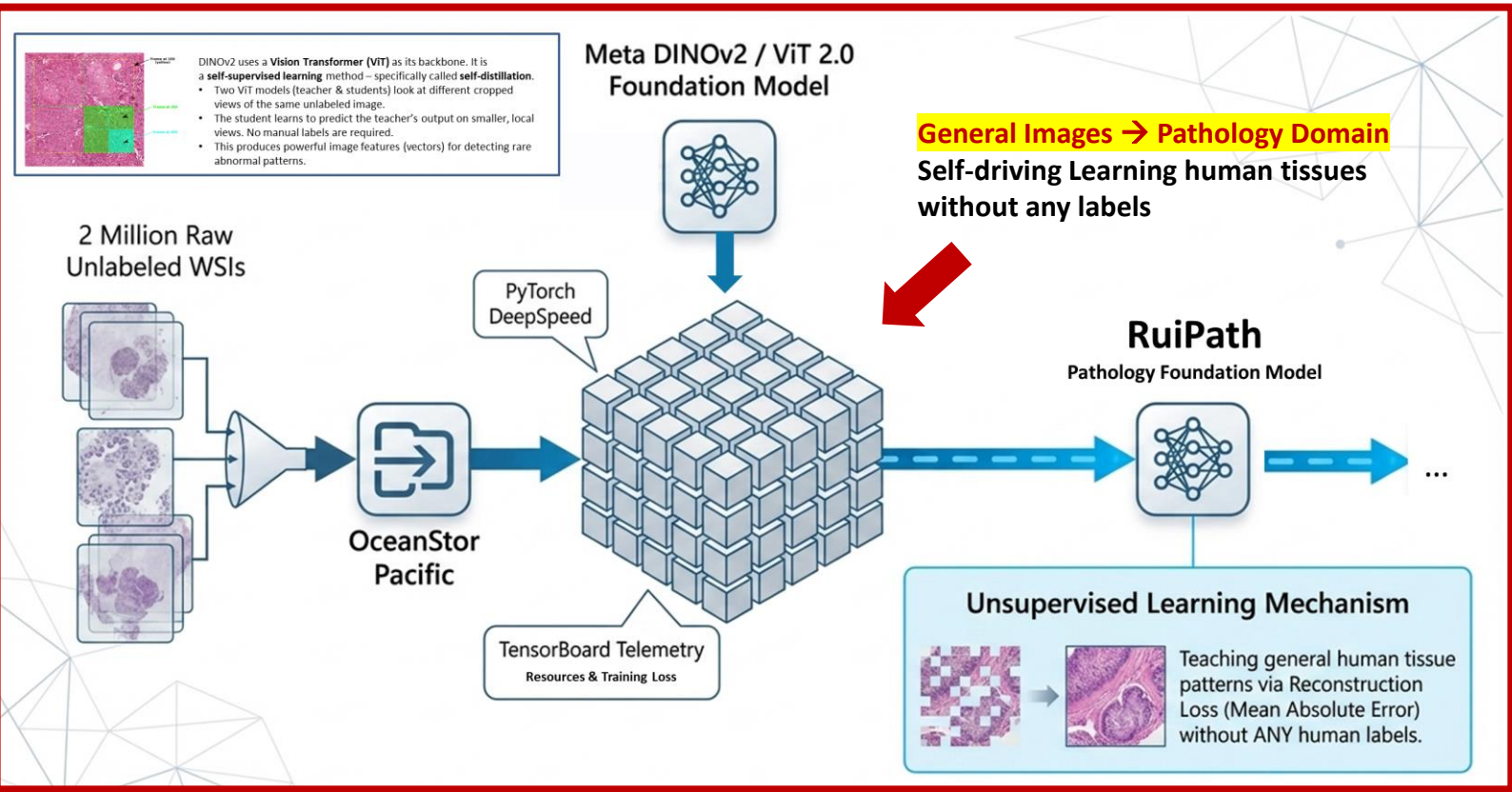
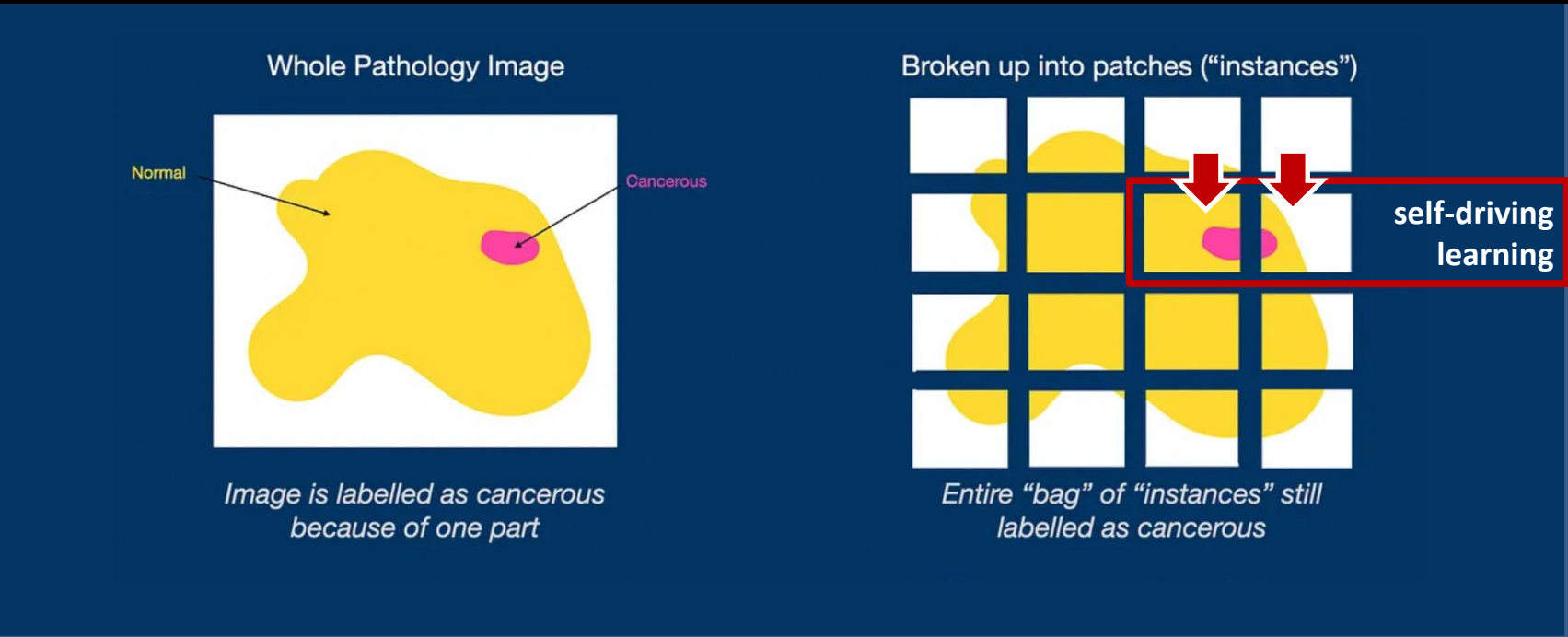
Vision Transformer **Scales** Digital Pathology

Database Modernization is about **building the AI data platform** around DBs

From AI assisted labelling to large scale self-learning

Modern Vision Transformer (ViT) empowers Multi-instance learning at Scale

Multi-instance learning (MIL) treats a whole slide image as a “bag” of smaller patches and, using only image-level labels (e.g., “contains tumor” or “normal”), **MIL can learn which patches are most indicative of tumor cells** — without requiring expensive cell-level annotations.



We fine-tuned a vision transformer using millions of whole slide images to build a foundation model for clinical pathology, make **diagnostic intelligence self-driving**

Feature	Legacy AI	Modern GenAI
Database Engine	X Single-instance PostgreSQL X Storing metadata and image path	☺ Distributed GaussDB (Relationship SQL + Vector Embedding) + OceanStor Pacific/S3
Analytics	X tradition AI/ML Dropped due to high cost to develop and week usage	☺ Easy to scale MIL models powered by VisionTransformer foundation model
Data State	X Archive only “dark data” X Unstructured inaccessible	☺ Vectorized Unstructured data for query and search, ready for analytics.
Clinical Workflow	X 100% Manual in digitalized computer screen & microscopic	☺ AI co-pilot with Image Diagnostic Agents & Clinical Knowledge Q&A (RAG)
Turnaround Time	X Multi-day Reporting	☺ 15 Mins end to end reporting Time

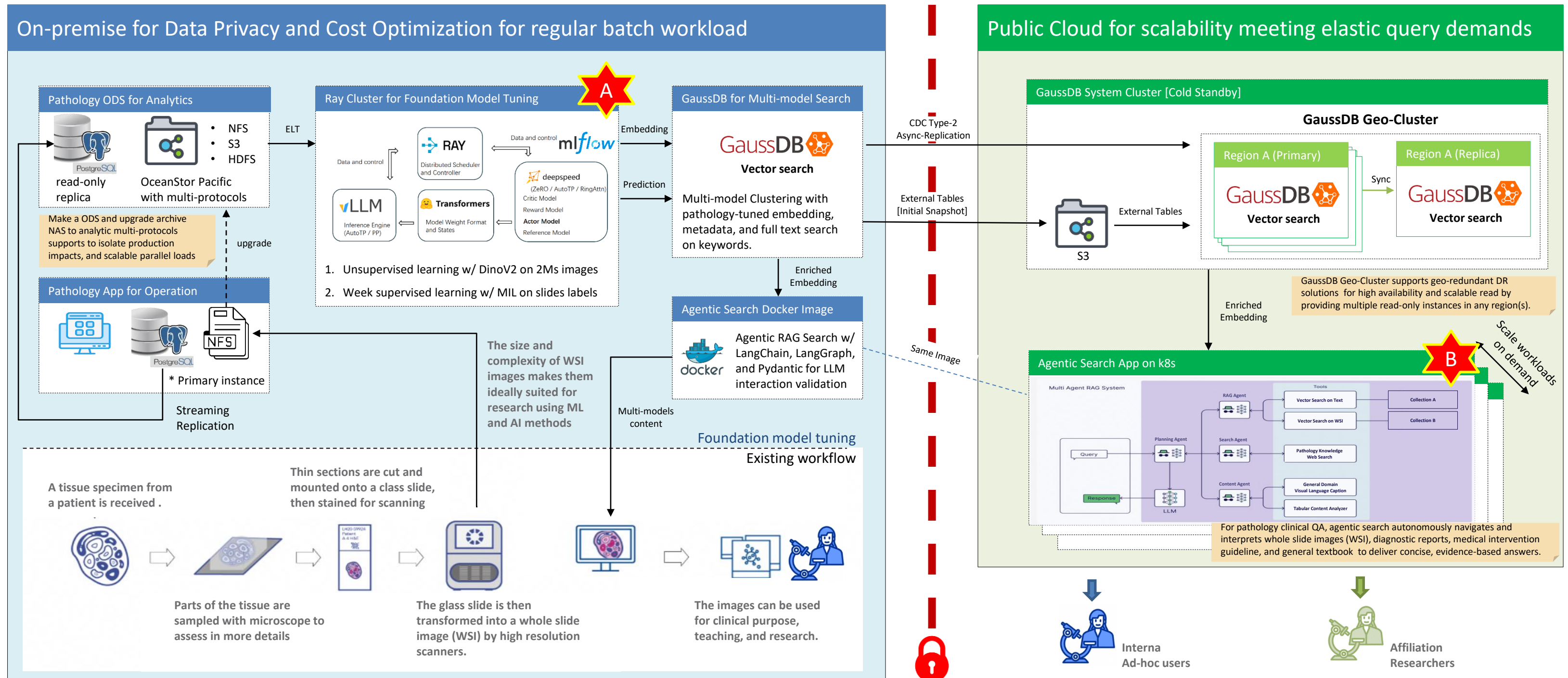
Architecture Overview - Hybrid Cloud Solution

Keeping Pathology Foundation Models and Patient Data on premise, and empowers Queries Scale at Cloud

Stable Workloads With Secure Data Boundary
(Regulatory Compliance – patient related data are not allowed in public cloud)



Scalable Workloads For Meeting Elastic Clinical Demands
(Dynamic Scalability - surge workloads at pandemic or public health crisis)

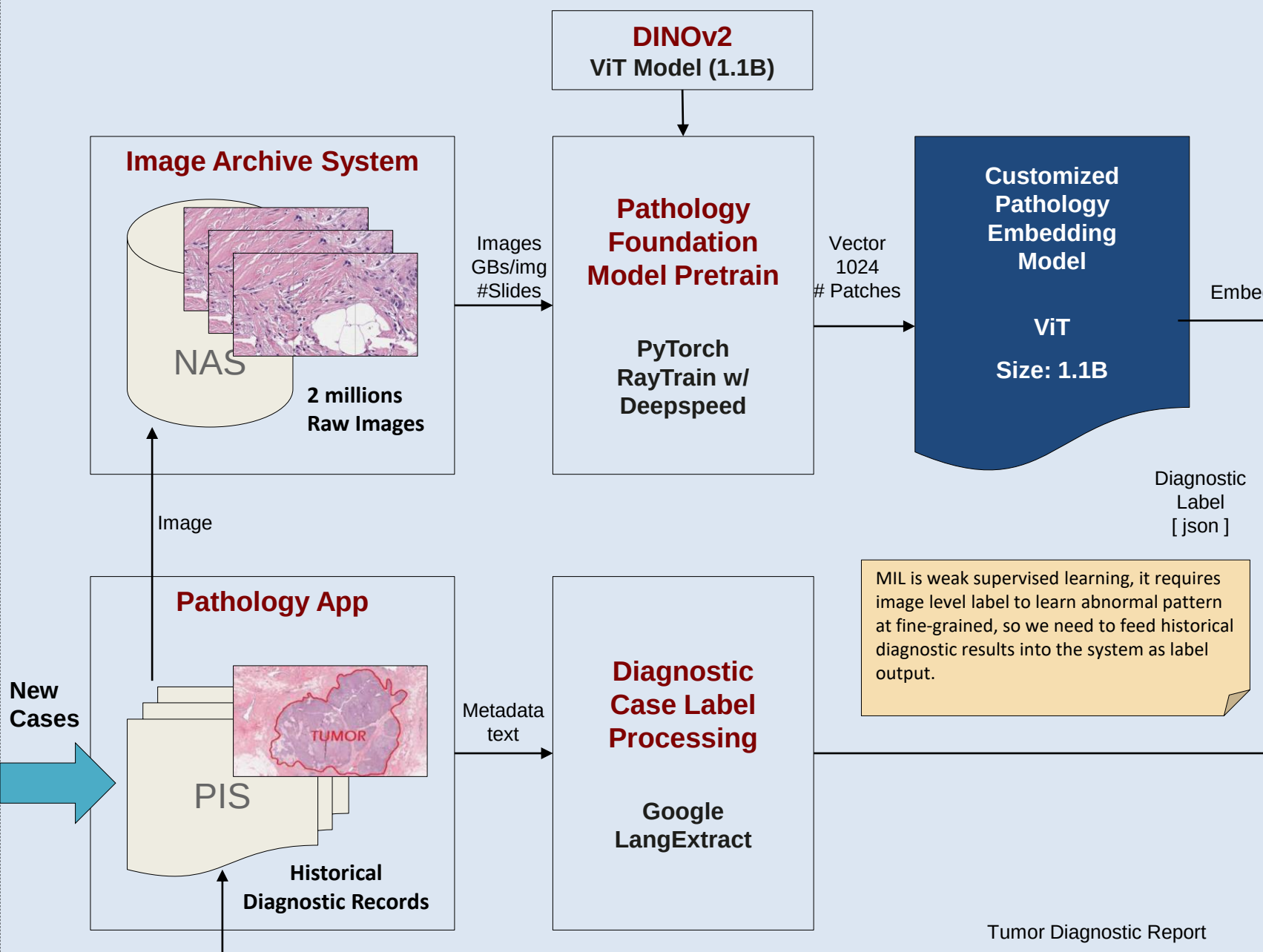


Core Tasks | Terms: { ODS: Operation Data Store; WSI: Whole Slide Image }

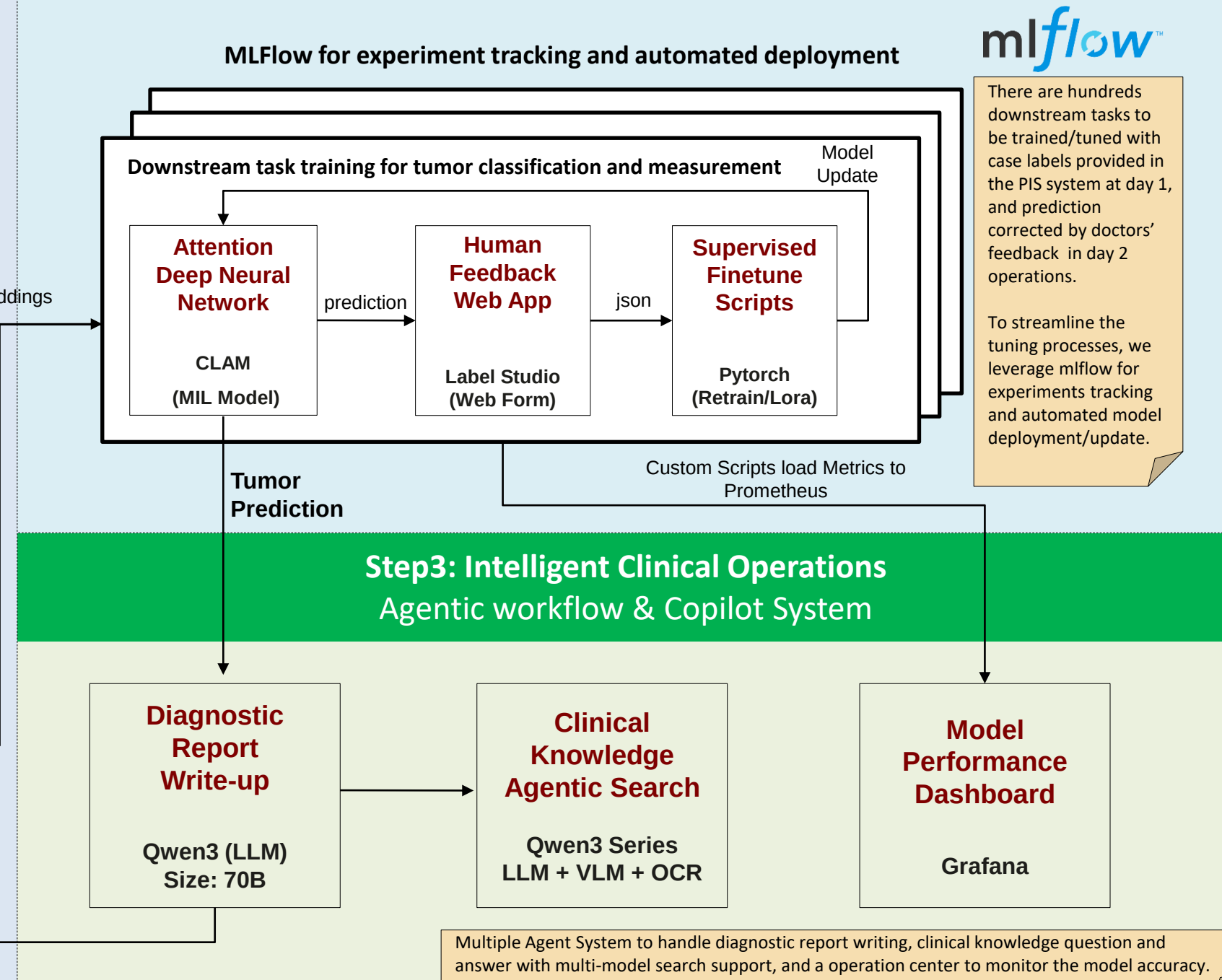
GenAI Core Task A: Model Development for Pathology Foundation Model

Modern Vision Transformer (ViT) empowers Multi-instance learning at Scale, learning image pattern with existing case labels

Step 1: Foundation Model Training (Once) Model Domain Adaptation System

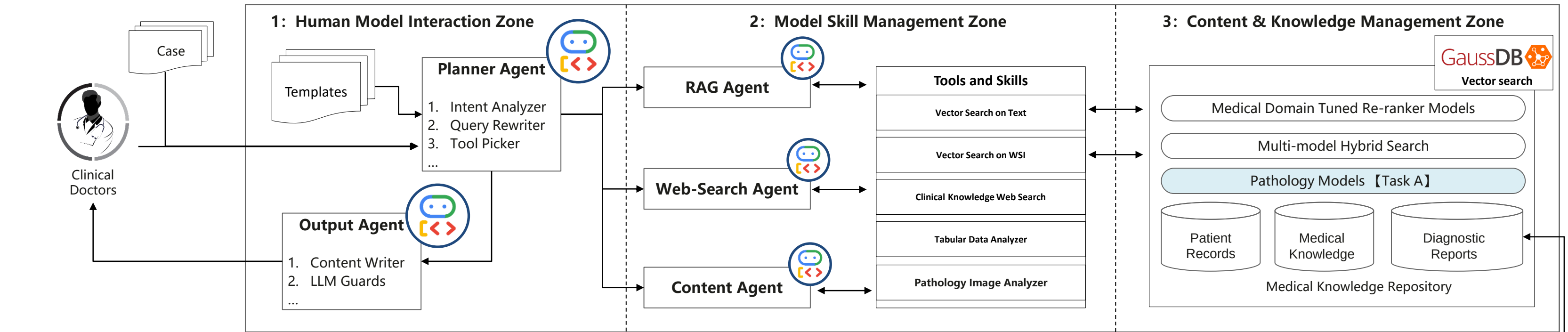


Step 2: Multi-instance Learning (Continuous) Feedback Learning System



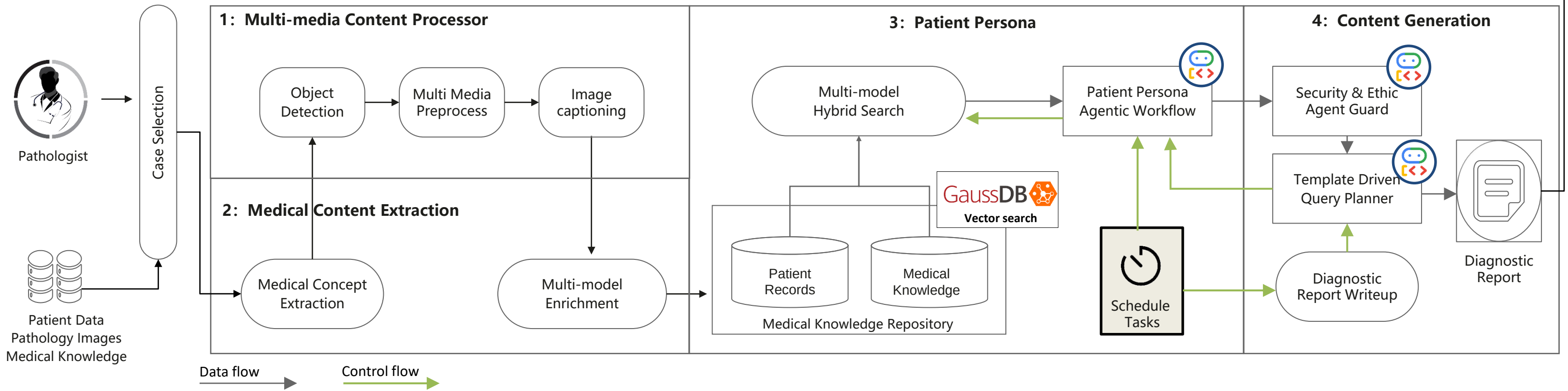
GenAI Core Task B: Multi-models Clinical Knowledge Q&A (Agentic RAG)

Agentic Search on Multi-modal data empowers accuracy contextual rich



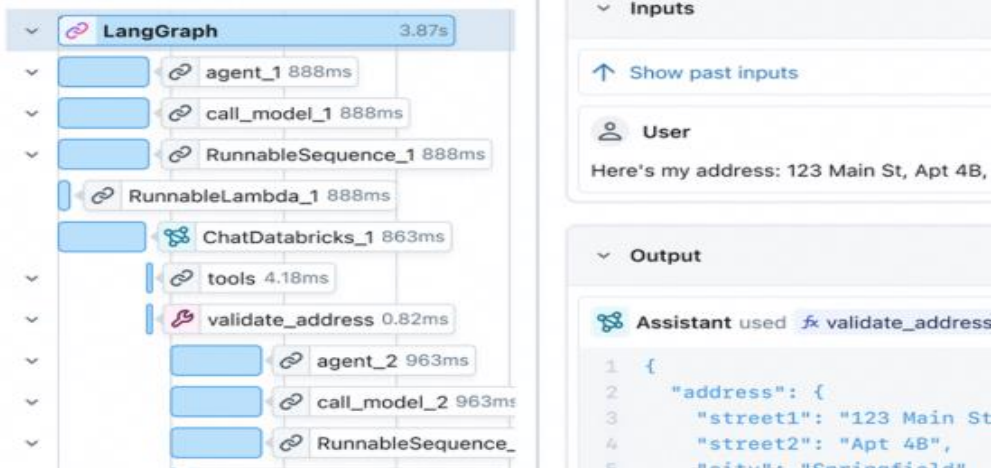
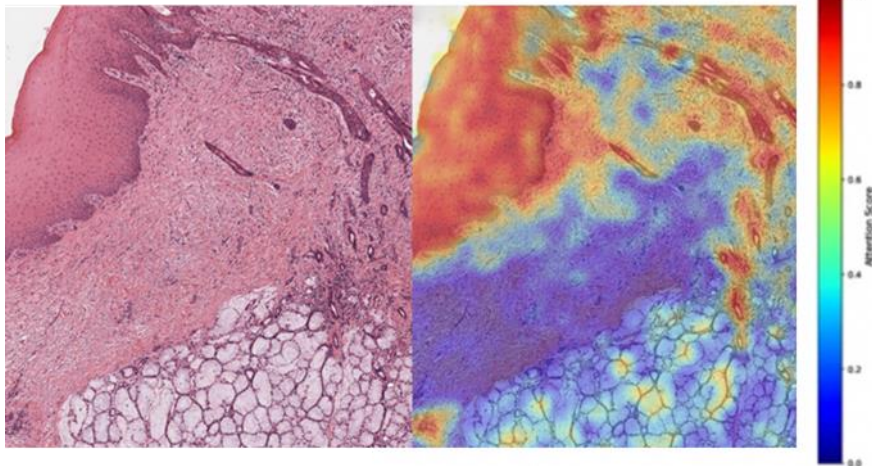
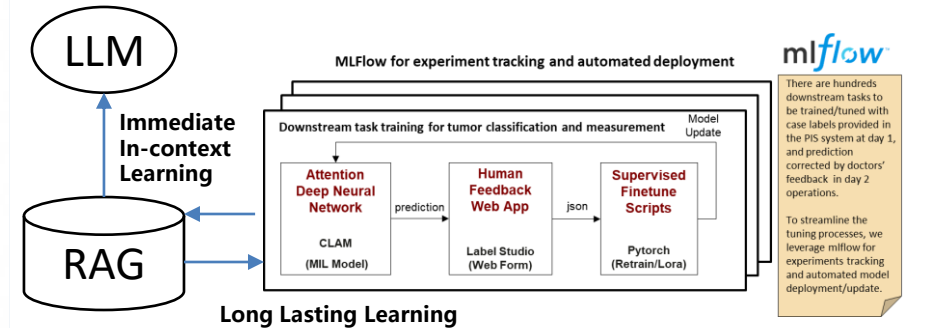
Agentic RAG Query

Data Ingestion & Schedule Jobs



Built-in Traceability, Explainability, and Rectifiability

Enterprise AI Platform Features for Regulated Workloads

	Traceability – Chain of Evidences	Explainability – Chain of Thoughts	Rectifiability – Chain of Feedbacks
What is it	Every AI output (e.g., “tumor detected”) can be traced back step by step to the original raw data that caused it (e.g. Pathology Images & Clinical Notes).	The AI inference shows its “reasoning” in human-understandable form – not just a score, but a sequence of logical steps, visual highlights, or textual justifications.	When a human expert corrects the AI’s mistake , that correction is captured and fed back into the system – updating the knowledge base, the agent workflow, and the model itself. The AI learns from its errors.
Why matters	Without traceability, you cannot audit or debug the AI. In regulated fields like healthcare or finance, regulators demand to know <i>why a decision was made</i> and <i>what data led to it</i> .	Regulated fields, like Clinicians, risk officers, or claims adjusters will not trust a black box. Explainability builds confidence, enables error detection, and helps <i>meet “right to explanation” regulations</i> .	No model is perfect. Without rectifiability, the same mistakes happen again and again. With it, the system <i>continuously improves</i> , reducing long-term maintenance cost and <i>building trust through collaboration</i> .
Solution Highlights	The model have to flag suspicious regions with metadata records, the platform provides: - With MLFlow model management, you have the transparency on the flow of sources → pipeline → model inference → Agent → output - With RAG reference sets, you know which documents and cases were retrieved 	The foundation models with attention scores, and 100s models for downstream tasks on classification and measurement provides: - With agentic workflow for chains-of-predictions which aligns with clinical practice, you know a clear and explainable composition to diagnostic prediction. - With attention heatmap, you know which part of the whole-slide image (WSI) likely contributed to the tumor. 	The AI platform provides two ways to learn from doctor feedback, a short term and a long term correction: Immediate correction – When a doctor corrects a mistake, we add the corrected image and label directly into the RAG knowledge base. This gives the LLM a real example to reference for similar future queries – like in-context learning. Long-term improvement – We use MLFlow to track and continuously fine-tune the vision transformer and prediction models using both positive and negative feedback samples. Over time, the model and agents gets smarter as it learns from more human corrections. 

Outcome

Measurable Business Impacts and Outstanding Research Output

Business Impacts



Production Status:

Deployed for Breast, Lung, and Lymph cancers.



Comprehensive Scope:

Tracking >70 distinct clinical metrics.

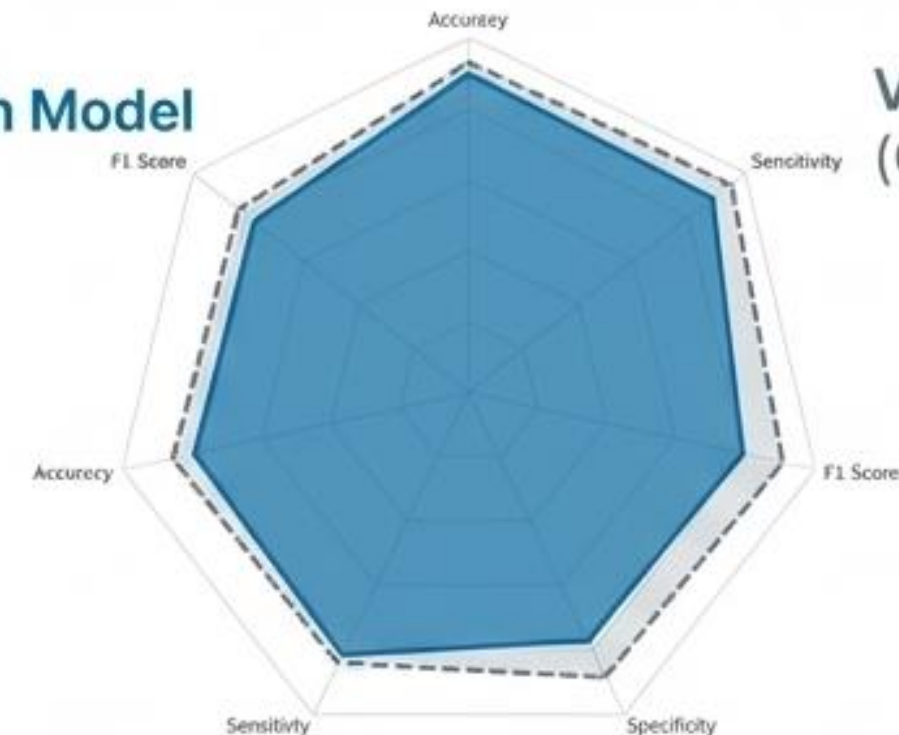


Clinical Precision:

Achieves >90% predictive accuracy.

Research Achievement

RuiPath Model



Virchow2 (Global SOTA)

- Evaluated against **12** mainstream public datasets.
- Tested across **14** auxiliary diagnostic tasks.

Result: 7 tasks achieved world-class performance (AUC, ACC, F1 score) exactly matching the global state-of-the-art Virchow2 model.

My Roles and Responsibilities

Engineering Leadership in High-Stakes Environments

AI Data Platform Solution Architect

- Designed the data platform (PyTorch + MLflow + RAG) with the **end-to-end data pipeline**: from raw S3 images → vectorization → training pipeline → real-time inference service.
- Led the agentic workflow system design with the hospital IT team and my engineering team on assuring **traceability, Explainability, and rectifiability** throughout every stages to ensure AI prediction is well governed.

Data & Model Engineers (Algorithms & Code)

- Built the solution with MIL model finetune + RAG integration – wrote the core **clustering algorithm, workflow for retrieval and ranking logic** that combines image vision features with clinical knowledge in text format.
- Established **MLFlow experiment tracking** and **automate model deployment** including hyperparameters, and evaluation metrics across hundreds of fine-tuning runs.

Agentic RAG Experts (Research Direction)

- **Agentic RAG workflow** – Multi-agent collaboration (retriever, verifier, rectifier) for complex reasoning tasks.
- **Multimodal retrieval** – Hybrid search combining vector similarity, keyword match, and knowledge graphs.
- **Auto-learning loop** – System improves continuously from both human corrections and AI-generated feedback.

Chiwai's Experience - Value Proposition

I bridge the gap between legacy data infrastructure and generative intelligence in heavily regulated industries—healthcare, finance, and public sector—by creating high-value use cases that turn dark data into operational assets, **making AI traceable, explainable, and rectifiable for production deployment**